

Design note: the Tier-3 ASTRA emulator for cosmological inference

ASTRA clustering project

June 2026

Abstract

The Fisher work (see the data-vector-search note) shows ASTRA quantile clustering can tighten ω_b, ω_c, n_s by several-fold over the 2PCF, but that the gain lives almost entirely at small scales ($s \lesssim 40 h^{-1}$ Mpc). Turning this into *measured* constraints needs a forward model $\xi(\theta)$ over the whole prior — an emulator trained on a space-filling cosmology grid — because a Fisher forecast only needs local derivatives, and small scales cannot be modelled analytically. This note records the Tier-3 design and the pilot.

1 What changes versus Fisher

Fisher needs derivatives at one fiducial, so the nine-cosmology derivative star suffices. Inference needs $\xi(\theta)$ everywhere in the prior, i.e. a space-filling cosmology grid plus an emulator. The catalogs exist: the source has **85 cosmologies** (c000–c181; we have used 9) and 500 HODs each (we have used 50). Tier-3 is therefore a compute problem, not a data one.

2 Design, driven by the findings

1. **Cosmologies: the space-filling block c130–c181** (~ 52 , the AbacusSummit emulator Latin hypercube). It varies the *broader* parameter set (w_0, w_a, N_{eff} , running), so the emulator inputs must include those (~ 8 -D cosmology), not just the four Λ CDM parameters. The nine derivative cosmologies are held out as validation anchors.
2. **HODs: ~ 100 – 150 per cosmology** from the unused 500. The HOD response is effectively ~ 4 -D (SIGMA, LOGM.CUT dominate) and 50 draws already captured $\sim 75\%$, so ~ 120 is ample. Total $\sim 52 \times 120 \approx 6000$ training runs.
3. **Focused, small-scale data vector.** The information is at $s \lesssim 40 h^{-1}$ Mpc and in the extreme-environment crosses (void/knot/filament, data and random). Emulate a reduced vector of those pieces, *finely binned at small scales* (~ 2 – $4 h^{-1}$ Mpc, vs the current $10 h^{-1}$ Mpc), not all 17 stems over 0–150.
4. **Phases.** Single phase (ph000) bakes in one cosmic-variance realisation — a problem at large scales, but negligible at the small scales that carry the signal — so ph000 is adequate for the emulator *mean*. Use the 25 AbacusSummit fiducial phases for the *covariance*.
5. **Iterations.** Run the full campaign at ~ 10 ASTRA iterations to lower the per-vector noise floor (the random legs benefit). The *pilot* uses 3 iterations (cheap, ~ 25 min, fits the 1-h queue limit) since the small scales are already signal-rich.

6. **Architecture.** ~ 6000 points over ~ 20 -D input $\Rightarrow$ an MLP (SUNBIRD-style), not a GP (which dies past a few-thousand points). PCA or direct multi-output, covariance-weighted MSE loss (matches the inference metric), an ensemble for the emulator uncertainty, and *leave-one-cosmology-out* validation (cosmology is the sparse axis).

3 The pilot (this note’s deliverable)

A small batch to decide go/no-go *before* the full campaign:

- **Runs:** 10 cosmologies spread over `c130–c181` $\times$ 50 maximin HODs = 500 full-box runs at 3 iterations, into the isolated root `data/fullbox_tier3/` (so the Fisher dataset is untouched). `scripts/select_tier3_pilot.py` + `queue/launch_tier3_pilot.sh`.
- **The one question it must answer:** can an emulator reach *sub-noise accuracy* at $s < 40 h^{-1}$ Mpc for the environment crosses, validated leave-one-cosmology-out? The entire Tier-3 payoff rides on this.
- **If yes:** run the full `c130–c181` $\times$ 120 campaign at 10 iterations, build the MLP, then MCMC on held-out mocks (does it recover truth; does the measured σ match the Fisher forecast — it will be looser, esp. σ_8), then apply to data.
- **If no:** fall back to the conservative-scale ($s > 40 h^{-1}$ Mpc) result, $\sim 1.5\times$ on ω_b, ω_c, n_s , with analytic modelling and no emulator.

4 Compute

Pilot: $500 \text{ runs} \times 3 \text{ iters} (\sim 25 \text{ min}) \approx 200 \text{ node-hours}$, backfills in $\sim 1\text{--}2$ days. Full campaign: $6000 \text{ runs} \times 10 \text{ iters} (\sim 1.5 \text{ h}) \approx 9000 \text{ node-hours}$ — a few weeks of queue, hence the pilot first.